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Reflection: Date Display GUI (HW13b)

1. Design Decisions and Rationale

To maximize modularity and maintainable code organization, the application strictly adheres to the **Model-View-Controller (MVC)** architectural pattern:

- **Model (`DateModel`)**: Encapsulates all temporal logic using `java.time.LocalDate`. It handles complex calendar edge cases, such as leap years and varying month lengths, inherently.
- **View (`DateView`)**: Manages the Java Swing GUI, displaying the date in the localized Vietnamese format (`dd/MM/yyyy`).
- **Controller (`DateController`)**: Acts as the intermediary, listening for user interactions, validating inputs, updating the Model, and requesting the View to refresh.

2. Main Functionalities

The application automatically retrieves and displays the current system date upon startup. Users can input an integer `x` and utilize four control buttons to navigate chronologically forward or backward by either days or months. A key feature is the integration of `JOptionPane` to provide immediate feedback, alerting users to input errors such as empty fields, non-numeric characters, or negative values.

3. Implementation Overview

The implementation focused heavily on input validation within the `Controller`. I employed a structured `try-catch` mechanism to detect `NumberFormatException` and throw custom `IllegalArgumentException` messages for invalid data. By leveraging the built-in `plusDays()` and `minusMonths()` methods of `LocalDate`, the application reliably addresses date-related edge cases, ensuring accuracy even when transitioning between months with different total days.